



2025 GCE O
level AM ...

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MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Ordinary Level



CANDIDATE NAME



CENTRE NUMBER

S

INDEX NUMBER

ADDITIONAL MATHEMATICS

Paper 1

Candidates answer on the Question Paper.
No Additional Materials are required.

4049/01

October/November 2025

2 hours 15 minutes

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE ON ANY BARCODES.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an approved scientific calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

This document consists of 16 printed pages.



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[Turn over]



1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}bc \sin A$$



- 1 An item is to be sold online. At the start of the month, it is placed on sale at an initial price of \$300. At the end of each month, if the item hasn't sold, the price is automatically reduced by 5%. It remains on sale at this price for the next month. The item finally sells in the month when the item is first priced below \$120.

How many complete months have elapsed before the item is sold?

[5]

$$300(0.95)^n < 120 \quad \text{--- MI}$$

$$(0.95)^n < \frac{120}{300}$$

$$n \ln(0.95) < \ln 0.4 \quad \text{--- MI}$$

$$n > \frac{\ln 0.4}{\ln 0.95}$$

$$n > 17.863 \quad \text{--- MI}$$

Complete months elapsed = 18 months --- AI

- 2 Find the set of values of the constant k for which the curve $y = kx^2 + 2x + k + 1$ is completely below the line $y = x + 1$. [6]

$$kx^2 + 2x + k + 1 = x + 1$$

$$kx^2 + x + k = 0 \quad \text{--- MI}$$

Discriminant < 0

$$(1)^2 - 4(k)(k) < 0 \quad \text{--- MI}$$

$$1 - 4k^2 < 0 \quad \text{--- MI}$$

$$(1+2k)(1-2k) < 0 \quad \text{--- MI}$$



$$k < -\frac{1}{2} \text{ or } k > \frac{1}{2} \text{ (rej, } k < 0) \quad \text{--- MI}$$

$$\therefore k < -\frac{1}{2} \quad \text{--- AI}$$

3 (a) (i) Factorise $a^3 - b^3$.

[1]

$$= (a-b)(a^2 + ab + b^2)$$

(ii) Hence state a factor, other than itself and 1, of $347^3 - 286^3$.

[1]

$$\text{Factor} = 347 - 286 = 61$$

It is given that x and n are positive integers.(b) Use part (a)(i) to prove that $\frac{(x+n)^3}{2} - \frac{(x-n)^3}{2}$ is divisible by n .

[3]

$$= \frac{1}{2} [(x+n)^3 - (x-n)^3]$$

$$= \frac{1}{2} [(x+n) - (x-n)] [(x+n)^2 + (x+n)(x-n) + (x-n)^2] \text{---M1}$$

$$= \frac{1}{2} (2n) [x^2 + 2nx + n^2 + x^2 - n^2 + x^2 - 2nx + n^2]$$

$$= n(3x^2 + n^2) \text{---M1}$$

M1 { Since x and n are positive integers, $3x^2 + n^2$ is also a positive integer.

Hence, $\frac{(x+n)^3}{2} - \frac{(x-n)^3}{2} = n(3x^2 + n^2)$ is divisible by n as it has n as a factor.



- 4 (a) Differentiate $x^{n+1} \ln x$ with respect to x , where $n \geq 0$.

[2]

$$\begin{aligned} \frac{d}{dx} [x^{n+1} \ln x] &= (\ln x)(n+1)x^n + x^{n+1} \left(\frac{1}{x}\right) \\ &= (n+1)\ln x \underbrace{x^n}_{A1} + \underbrace{x^n}_{A1} \\ &= [(n+1)\ln x + 1] x^n \end{aligned}$$

- (b) Hence find $\int x^n \ln x dx$.

[3]

From (a), $\int [(n+1)\ln x + 1] x^n dx = x^{n+1} \ln x + C$

$$(n+1) \int x^n \ln x dx + \int x^n dx = x^{n+1} \ln x + C \quad -M1$$

$$(n+1) \int x^n \ln x dx + \frac{x^{n+1}}{n+1} = x^{n+1} \ln x + C \quad -M1$$

$$\begin{aligned} \int x^n \ln x dx &= \frac{x^{n+1} \ln x}{n+1} - \frac{x^{n+1}}{(n+1)^2} + \frac{C}{n+1} \\ &= \frac{x^{n+1} \ln x}{n+1} - \frac{x^{n+1}}{(n+1)^2} + C' \quad -A1 \end{aligned}$$

- 5 Two variables x and y are believed to be related by the equation $y = Ax^b$ where A and b are constants. When a set of points (X, Y) are plotted, where X is a function of x and Y is a function of y , the resulting graph is a straight line.

- (a) Given that $\ln y$ is plotted on the Y -axis determine, with working, the function of x which must be plotted on the X -axis. [2]

$$\begin{aligned}\ln y &= \ln(Ax^b) \\ &= \ln A + \ln(x^b) \\ &= b \ln x + \ln A - M1 \\ \text{Function of } x &= \ln x. - A1\end{aligned}$$

- (b) Given that two of the points lying on the straight line are $(1.2, 6.3)$ and $(4.1, 12.1)$ find the values of A and b . [4]

$$\ln y = b \ln x + \ln A$$

$$\begin{aligned}b &= \text{gradient} \\ &= \frac{12.1 - 6.3}{4.1 - 1.2} - M1 \\ &= 2. - A1\end{aligned}$$

$$\begin{aligned}\text{Sub } (1.2, 6.3): \quad 6.3 &= 2(1.2) + \ln A - M1 \\ \ln A &= 3.9 \\ A &= e^{3.9} \\ &= 49.4 \text{ (3sf)} - A1\end{aligned}$$

- 6 (a) Show that the x -coordinate of the stationary point on the curve $y = \frac{1}{2}x^2 - 3\sqrt{x} + 1$ is $\left(\frac{3}{2}\right)^{\frac{2}{3}}$. [4]

$$\begin{aligned}\frac{dy}{dx} &= x - \frac{3}{2}x^{-\frac{1}{2}} \\ \text{when } \frac{dy}{dx} &= 0, \quad x - \frac{3}{2\sqrt{x}} = 0. \quad -M1 \\ x &= \frac{3}{2\sqrt{x}} \\ 2x\sqrt{x} &= 3. \\ x^{1.5} &= \frac{3}{2} \\ x &= \left(\frac{3}{2}\right)^{\frac{2}{3}}. \quad -A1\end{aligned}$$

- (b) Determine the nature of this stationary point. [3]

1st derivative test:

	$\left(\frac{3}{2}\right)^{\frac{2}{3}} = 1.3$	$\left(\frac{3}{2}\right)^{\frac{2}{3}} = 1.31$	$\left(\frac{3}{2}\right)^{\frac{2}{3}} = 1.32$
$\frac{dy}{dx}$	-0.0155 (< 0)	0	0.0144 (> 0)
Sketch			

stationary point is a minimum. -A1

OR

$$\begin{aligned}\frac{d^2y}{dx^2} &= 1 + \frac{3}{4}x^{-1.5} \quad -M1 \\ \text{when } x &= \left(\frac{3}{2}\right)^{\frac{2}{3}}, \quad \frac{d^2y}{dx^2} = 1.5 > 0 \quad -M1\end{aligned}$$

Hence, stationary point is a minimum point. -A1

- 7 (a) Find, in ascending powers of x , the first four terms in the expansion of $\left(\frac{1}{2} - ax\right)^5$, where a is a constant. Simplify the coefficients in your expansion. [4]

$$\begin{aligned}\left(\frac{1}{2} - ax\right)^5 &= \binom{5}{0}\left(\frac{1}{2}\right)^5 + \binom{5}{1}\left(\frac{1}{2}\right)^4(-ax) + \binom{5}{2}\left(\frac{1}{2}\right)^3(-ax)^2 + \binom{5}{3}\left(\frac{1}{2}\right)^2(-ax)^3 + \dots \\ &= \frac{1}{32} - \frac{5ax}{16} + \frac{5a^2x^2}{4} - \frac{5a^3x^3}{2} + \dots\end{aligned}$$

(work for each term.)

- (b) In the expansion of $\left(4 + \frac{5}{ax}\right)\left(\frac{1}{2} - ax\right)^5$, find the non-zero value of a for which the coefficient of x is twice the coefficient of x^2 . [3]

$$\left(4 + \frac{5}{ax}\right)\left(\frac{1}{32} - \frac{5ax}{16} + \frac{5a^2x^2}{4} - \frac{5a^3x^3}{2} + \dots\right)$$

$$\text{coefficient of } x = (4)\left(-\frac{5a}{16}\right) + \left(\frac{5}{a}\right)\left(\frac{5a^2}{4}\right)$$

$$= -\frac{5a}{4} + \frac{25a}{4}$$

$$= 5a$$

$$\text{coefficient of } x^2 = (4)\left(\frac{5a^2}{4}\right) + \left(\frac{5}{a}\right)\left(-\frac{5a^3}{2}\right)$$

$$= 5a^2 - \frac{25a^2}{2}$$

$$= -\frac{15a^2}{2}$$

$$5a = 2\left(-\frac{15a^2}{2}\right) \quad -M1$$

$$5a = -15a^2$$

$$5a + 15a^2 = 0$$

$$5a(1 + 3a) = 0$$

$$a = 0 \text{ or } a = -\frac{1}{3}$$

$$(not) \therefore a = -\frac{1}{3} \quad -A1$$

- 8 A curve has equation $\frac{(x-a)^2}{2} + (y-1)^2 = 1$, where a is a constant.

(a) Find the set of values of a for which the line $y = x+1$ intersects this curve at two distinct points. [5]

$$\frac{(x-a)^2}{2} + (x+1-1)^2 = 1$$

$$\frac{(x-a)^2}{2} + x^2 = 1$$

$$(x-a)^2 + 2x^2 = 2$$

$$x^2 - 2ax + a^2 + 2x^2 - 2 = 0$$

$$3x^2 - 2ax + a^2 - 2 = 0 \quad -M1$$

$$\text{Discriminant} > 0.$$

$$(-2a)^2 - 4(3)(a^2 - 2) > 0 \quad -M1$$

$$4a^2 - 12a^2 + 24 > 0.$$

$$-8a^2 + 24 > 0.$$

$$a^2 - 3 < 0.$$

$$(a+\sqrt{3})(a-\sqrt{3}) < 0. \quad -M1$$

$$\frac{a+\sqrt{3}}{-\sqrt{3}} \frac{a-\sqrt{3}}{\sqrt{3}}$$

$$a < -\sqrt{3} \text{ or } a > \sqrt{3} \quad -A1$$

$$(\text{or } a < -1.73 \text{ or } a > 1.73)$$

(b) Deduce the two values of a for which the line $y = x+1$ is a tangent to the curve. Show your working clearly. [2]

$$\text{Discriminant} = 0.$$

$$(a+\sqrt{3})(a-\sqrt{3}) = 0 \quad -M1$$

$$a = \sqrt{3} \text{ or } -\sqrt{3}$$

$$A1$$

$$(\text{or } a = \pm 1.73)$$





9 (a) Prove that $\frac{1+\sin 2\theta}{\sin \theta} - \frac{1+\cos 2\theta}{\cos \theta} = \operatorname{cosec} \theta$.

[4]

$$\begin{aligned}
 \text{LHS} &= \frac{1+\sin 2\theta}{\sin \theta} - \frac{1+\cos 2\theta}{\cos \theta} \\
 &= \frac{1}{\sin \theta} + \frac{2\sin \theta \cos \theta}{\sin \theta} - \frac{(1+2\cos^2 \theta - 1)}{\cos \theta} \quad \text{--- M1} \\
 &= \operatorname{cosec} \theta + 2\cos \theta - \frac{2\cos^2 \theta}{\cos \theta} \\
 &= \operatorname{cosec} \theta + 2\cos \theta - 2\cos \theta \quad \text{--- M1} \\
 &= \operatorname{cosec} \theta \\
 &= \text{RHS.} \quad \text{--- A1}
 \end{aligned}$$

(b) Hence solve $\frac{1+\sin 2\theta}{\sin \theta} - \frac{1+\cos 2\theta}{\cos \theta} = 2\sin \theta$ for $\pi < \theta < \frac{3}{2}\pi$.

[3]

$$\begin{aligned}
 \operatorname{cosec} \theta &= 2\sin \theta \\
 \frac{1}{\sin \theta} &= 2\sin \theta \\
 2\sin^2 \theta &= 1 \\
 \sin^2 \theta &= \frac{1}{2} \\
 \sin \theta &= \frac{1}{\sqrt{2}} \text{ or } -\frac{1}{\sqrt{2}} \quad \text{--- M1} \\
 \text{heuristic angle} &= \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4} \quad \text{--- M1} \\
 \text{since } \theta \text{ is in 3rd quadrant,} \\
 \theta &= \pi + \frac{\pi}{4} \\
 &= \frac{5\pi}{4} \quad \text{--- A1}
 \end{aligned}$$



- 10 (a) Find, in terms of the positive constant a , the equation of the normal to the curve $y = 2 \sin ax - 3 \cos 2x$ at the point P where $x = 0$. [7]

$$\frac{dy}{dx} = 2a \cos ax + 6 \sin 2x$$

$$\text{when } x=0, \frac{dy}{dx} = 2a - 6$$

$$\text{gradient of normal} = -\frac{1}{2a} - 6$$

$$y = -3 - 6$$

$$\text{gradient of normal: } \frac{y-3}{x-0} = -\frac{1}{2a} - 6$$

$$y = -\frac{1}{2a}x + 3 - 6$$

- (b) The normal cuts the x -axis at Q . The area of triangle OPQ , where O is the origin, is 27 units². Find the value of a . [3]

$$\text{when } y=0, 0 = -\frac{1}{2a}x + 3$$

$$\frac{1}{2a}x = 3$$

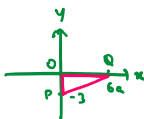
$$x = 6a - 6$$

$$P(0, -3); Q(6a, 0)$$

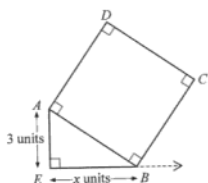
$$\frac{1}{2} \times 3 \times 6a = 27 - 6$$

$$6a = 18$$

$$a = 3 - 6$$



11



A square $ABCD$ is part of a computer animation. The side AB is the hypotenuse of a right-angled triangle AEB with angle $AEB = 90^\circ$. The points A and E are fixed and do not change during the animation.

The length of AE is always 3 units and angle AEB is always 90° .

During the animation B moves directly away from E along the line perpendicular to AE and, at time t seconds, the distance EB is x units.

It is given that x is increasing at a constant rate of 2 units/s.

- (a) Find the rate of change of the length AB , in units/s, when $x = 4$.

[5]

$$\text{length of } AB, L = \sqrt{3^2 + x^2} = \sqrt{x^2 + 9} \quad -M1$$

$$\text{Have: } \frac{dx}{dt}, \text{ to find: } \frac{dL}{dt}, \text{ need: } \frac{dL}{dx}.$$

$$\frac{dL}{dx} = \frac{1}{2}(x^2 + 9)^{-\frac{1}{2}}(2x) \quad -M1$$

$$\text{when } x = 4, \frac{dL}{dx} = \frac{1}{2}(25)^{-\frac{1}{2}}(8) = 0.8 \quad -M1$$

$$\frac{dL}{dt} = \frac{dL}{dx} \times \frac{dx}{dt}$$

$$= 0.8 \times 2 \quad -M1$$

$$= 1.6 \text{ units/s.} \quad -A1$$



- (b) The rate of change of the area of the square cannot exceed 80 units²/s. Find the maximum possible value for x . [3]

$$\text{Area, } A = x^2 + 9$$

$$\frac{dA}{dx} = 2x \quad - M1$$

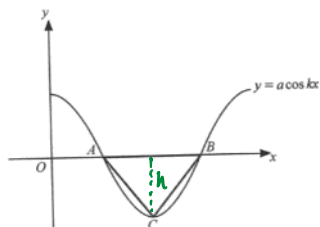
$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$= 4x \text{ units}^2/\text{s.} \quad - M1$$

$$4x \leq 80$$

$$x \leq 20.$$

$$\text{maximum } x = 20. \quad - A1$$



The diagram shows part of the curve $y = a \cos kx$ where a and k are positive constants and x is in radians. The curve crosses the positive x -axis at A and B .

- (a) (i) Find, in terms of k , the x -coordinates of A and of B .

[2]

$$\text{one period} = \frac{2\pi}{k}$$

$$x_A = \frac{2\pi}{k} \div 4 = \frac{\pi}{2k} \quad -A1$$

$$x_B = \frac{2\pi}{k} \times \frac{3}{4} = \frac{3\pi}{2k} \quad -A1$$

Point C is the minimum point on the curve between A and B .
Triangle ABC is equilateral.

- (ii) Prove that $a = \frac{\pi\sqrt{3}}{2k}$.

[3]

$$x_C = \frac{2\pi}{k} \times \frac{1}{2} = \frac{\pi}{k}, \quad y_C = -a$$

$$AB = \frac{3\pi}{2k} - \frac{\pi}{2k} = \frac{\pi}{k}$$

$$AC = \sqrt{\left(\frac{\pi}{k} - \frac{\pi}{2k}\right)^2 + (-a - 0)^2} \quad -M1$$

$$\sqrt{\frac{\pi^2}{4k^2} + a^2} = \frac{\pi}{k} \quad -M1$$

$$\frac{\pi^2}{4k^2} + a^2 = \frac{\pi^2}{k^2}$$

$$a^2 = \frac{3\pi^2}{4k^2}$$

$$a = \frac{\sqrt{3}\pi}{2k} \text{ or } -\frac{\sqrt{3}\pi}{2k} \text{ (reject, } a > 0)$$

$$\therefore a = \frac{\pi\sqrt{3}}{2k} \quad -A1$$



- (b) Given that the area of triangle ABC is 1 unit², show that $k = \frac{(3)^{\frac{1}{4}}\pi}{2}$. Hence find the amplitude and period of the curve. [3]

$$\frac{1}{2} \times \frac{\pi}{k} \times a = 1$$

$$\frac{\pi}{2k} \times \frac{\pi\sqrt{3}}{2k} = 1$$

$$\pi^2\sqrt{3} = 4k^2$$

$$k^2 = \frac{\pi^2\sqrt{3}}{4}$$

$$k = \frac{3^{\frac{1}{4}}\pi}{2} \approx \frac{3^{\frac{1}{4}}\pi}{2} \text{ (note } k > 0) \text{ — A1}$$

$$\text{amplitude } a = \frac{\pi\sqrt{3}}{2\left(\frac{3^{\frac{1}{4}}\pi}{2}\right)} = 3^{\frac{1}{4}} = 1.32 \text{ — A1}$$

$$\text{period} = \frac{2\pi}{k} = \frac{2\pi}{\left(\frac{3^{\frac{1}{4}}\pi}{2}\right)} = 3.04 \text{ — A1}$$



Question 13 is printed on the next page.



- 13 Points $A(-1, 7)$ and B lie on a circle.
 The chord AB of this circle has equation $2y - 3x = 17$.
 The centre of the circle lies on the line $y = -2$.
 The diameter of the circle parallel to AB passes through the point $(-\frac{1}{2}, \frac{1}{4})$.

(a) Find the equation of the circle.

Let centre of circle C be $(c, -2)$.

$$2y - 3x = 17$$

$$2y = 3x + 17$$

$$y = 1.5x + 8.5 \quad -M1$$

$$M1 - \left[\frac{-2 - \frac{1}{4}}{c - (-\frac{1}{2})} \right] = 1.5 - M1$$

$$-2.25 = 1.5c + 0.75$$

$$-3 = 1.5c$$

$$c = -2. \quad -M1$$

$$r^2 = AC^2 = (-2 - (-1))^2 + (-2 - 7)^2 = 82$$

$$\text{equation of circle: } (x+2)^2 + (y+2)^2 = 82.$$

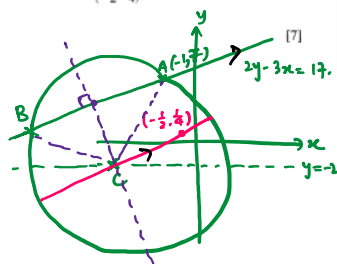
(b) The point $(7, \alpha)$ lies inside the circle. Find the set of possible values for α .

$$(7 - (-2))^2 + (\alpha - 2)^2 < 82. \quad -M1$$

$$(\alpha - 2)^2 < 1.$$

$$-1 < \alpha - 2 < 1$$

$$1 < \alpha < 3. \quad -A1$$



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